



Name: _____ Cohort/Batch: _____ Date: _____ Score: _____

ECHO PRACTICE SHEET - 10 CONCEPTUAL Q&A

10 Q&A on routing, middleware, binding, JWT auth, and testing

TOTAL: 10 QUESTIONS

Q1 What is Echo and how does it compare to Gin?

Threat Model

Q6 What does echo.Context provide for reading request data?

Q2 How do you apply middleware globally vs to a route group in Echo?

Q7 How do you implement JWT authentication in Echo?

Lifecycle

Q3 How does request binding and validation work in Echo?

Configuration

Q8 What are Echo's response helpers and what do they return?

Q4 How do Echo Groups help organize routes?

Hardening

Q9 How do you handle WebSocket connections in Echo?

Testing

Q5 How does Echo's custom error handler work?

SSO / IdP

Q10 How do you test Echo handlers without starting an HTTP server?

Tuning



ANSWER KEY & EXPLANATORY SOLUTIONS

Comprehensive verified answers, best-practice configs, security controls & reference

VERIFIED SOLUTIONS

A1 Both use radix-tree routing. Echo provides built-in

Mitigation

Both use radix-tree routing. Echo provides built-in middleware (Logger, Recover, CORS, JWT), automatic TLS via Let's Encrypt, and HTTP/2 support out of the box. Its echo.Context API is arguably more ergonomic with typed header/cookie helpers.

A2 e.Use(middleware.Logger()) applies globally to all routes

Sanitizes

e.Use(middleware.Logger()) applies globally to all routes. g := e.Group('/admin', AdminMiddleware) applies to the group only. Individual routes: e.GET('/profile', handler, AuthMiddleware). Middleware functions have signature func(next echo.HandlerFunc) echo.HandlerFunc.

A3 c.Bind(&req) reads JSON body, form values, or

Hardening

c.Bind(&req) reads JSON body, form values, or query params into a struct based on tags. Echo does not validate by default. Register a validator (e.Validator = &MyValidator{}) that implements Validate(i interface{}) error and call c.Validate(&req) explicitly.

A4 g := e.Group('/api/v1') prefixes all routes in

Gotchas

g := e.Group('/api/v1') prefixes all routes in the group. Apply shared middleware once: g.Use(JWTMiddleware). Create sub-groups: admin := g.Group('/admin', AdminOnly). This mirrors Express.Router and NestJS Module organization patterns.

A5 Set e.HTTPErrorHandler = func(err error, c echo.Context)

Centralization

Set e.HTTPErrorHandler = func(err error, c echo.Context). Check if err is *echo.HTTPError for status/message; otherwise default to 500. This centralizes error formatting, allowing you to return consistent JSON error shapes across the entire application.

A6 c.Param('id') reads path params, c.QueryParam('q') reads

Sanitization

c.Param('id') reads path params, c.QueryParam('q') reads query strings, c.FormValue('field') reads form fields, c.Request().Header.Get('X-API-Key') reads headers, and c.Get/Set passes values through middleware to handlers via the request context.

A7 Register echojwt.WithConfig(echojwt.Config{SigningKey: []byte(secret)})

Sanitization

Register echojwt.WithConfig(echojwt.Config{SigningKey: []byte(secret)}) as middleware on a Group. It reads the Bearer token, validates the signature, and populates the jwt.Token in the context. Retrieve claims with c.Get('user').(*jwt.Token).Claims.

A8 c.JSON(200, obj) marshals obj and sets Content-Type

Governance

c.JSON(200, obj) marshals obj and sets Content-Type application/json. c.XML, c.String, c.HTML, c.File, c.Stream cover other content types. c.NoContent(204) sends an empty response. c.Redirect(301, url) sets Location header and status.

A9 Use github.com/gorilla/websocket inside an Echo handler

Sanitization

Use github.com/gorilla/websocket inside an Echo handler. Register a standard echo.HandlerFunc, upgrade the connection with upgrader.Upgrade(c.Response(), c.Request(), nil), then loop on conn.ReadMessage() and conn.WriteMessage() for full-duplex communication.

A10 req := httptest.NewRequest(http.MethodGet, 'users/1', nil); rec :=

Sanitization

req := httptest.NewRequest(http.MethodGet, 'users/1', nil); rec := httptest.NewRecorder(); c := e.NewContext(req, rec); handler(c). Access rec.Code and rec.Body.String() for assertions. No listening port is required.